WEEK 3 :TASK BY: RIDDHI MALHOTRA

SQL

--DATABASE:

--A database is an organized collection of data that can be easily accessed,managed, and updated.

--It is designed to store information in a structured way so that users and applications can retrieve and manipulate it efficiently.

--TERMS OF DATABASE:

--Tables: Data is stored in tables, which are made up of rows (records) and columns (fields).

--Records (Rows): Each row represents a single entry in the table.

--Fields (Columns): Each column represents a specific attribute of the data.

--IMPORTANCE OF DATABASE:

--structured

--consistent

--integrity

--scalability

```
Create database InternNova_Riddhi_Malhotra
--In SQL, you use the CREATE DATABASE command to make a new database.

use [InternNova_Riddhi_Malhotra]
--After creating it, you need to tell SQL which database you're working with.

CREATE TABLE Employees (
   EmpID INT PRIMARY KEY,          -- Unique employee ID
   Name VARCHAR(50),               -- Employee name
   Department VARCHAR(30),         -- Department name
   Salary DECIMAL(10,2)            -- Salary with 2 decimal places
);

--Insert sample records
INSERT INTO Employees (EmpID, Name, Department, Salary)
VALUES
(101, 'Niklaus', 'HR', 50000),
(102, 'Elijah', 'IT', 60000),
(103, 'Rebekah', 'Finance', 55000),
(104, 'Hayley', 'IT', 58000);
```

```sql
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use [InternNova_Riddhi_Malhotra]
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VALUES
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(102, 'Elijah', 'IT', 60000),
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(104, 'Hayley', 'IT', 58000);
```

----------TASK 1----------

--1 SQL query to display all the records

select * from Employees

```sql
--1 SQL query to display all the records
select * from Employees
```

90 % No issues found

Results Messages

	EmpID	Name	Department	Salary
1	101	Niklaus	HR	50000.00
2	102	Elijah	IT	60000.00
3	103	Rebekah	Finance	55000.00
4	104	Hayley	IT	58000.00

--2 select specific column

select EmpID from Employees

select Name from Employees

select Department from Employees

select Salary from Employees

```
42
43    --2 select specific column
44    select EmpID from Employees
45    select Name from Employees
46    select Department from Employees
47    select Salary from Employees
```

90 % No issues found

Results Messages

	EmpID
1	101
2	102
3	103
4	104

	Name
1	Niklaus
2	Elijah
3	Rebekah
4	Hayley

	Department
1	HR
2	IT
3	Finance
4	IT

	Salary
1	50000.00
2	60000.00
3	55000.00
4	58000.00

--3 use allias where required

SELECT Name AS EmployeeName, Salary AS MonthlySalary

FROM Employees;

```
48
49    --3 use allias where required
50    SELECT Name AS EmployeeName, Salary AS MonthlySalary
51    FROM Employees;
52
53
54    ---Task 1 introduces the foundation of databases and
```

90 % No issues found

Results Messages

	EmployeeName	MonthlySalary
1	Niklaus	50000.00
2	Elijah	60000.00
3	Rebekah	55000.00
4	Hayley	58000.00

---Task 1 introduces the foundation of databases and demonstrates how the SELECT statement is used to query data.

--It shows how to retrieve all records, specific columns, and how to make results more readable with aliases.

----------TASK 2----------

--1 using the where clause

SELECT *

FROM Employees

WHERE Department = 'IT';

--2using comparison operator

SELECT *

FROM Employees

WHERE Salary > 55000;

```
57  ----------TASK 2-----
58  --1 using the where clause
59  SELECT *
60  FROM Employees
61  WHERE Department = 'IT';
62
63  --2using comparison operator
64  SELECT *
65  FROM Employees
66  WHERE Salary > 55000;
67
```

90 % No issues found

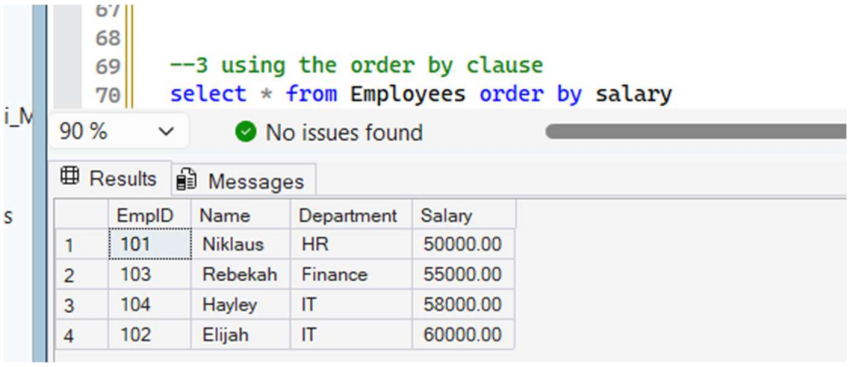
Results Messages

	EmpID	Name	Department	Salary
1	102	Elijah	IT	60000.00
2	104	Hayley	IT	58000.00

	EmpID	Name	Department	Salary
1	102	Elijah	IT	60000.00
2	104	Hayley	IT	58000.00

--3 using the order by clause

select * from Employees order by salary



	EmpID	Name	Department	Salary
1	101	Niklaus	HR	50000.00
2	103	Rebekah	Finance	55000.00
3	104	Hayley	IT	58000.00
4	102	Elijah	IT	60000.00

--4 aggregate functions:

--count():to find the total number of records

SELECT COUNT(*) AS TotalEmployees

FROM Employees;

--**sum():to calculate the total salary.**

SELECT SUM(Salary) AS TotalSalary

FROM Employees;

--**avg():to calculate the average salary.**

SELECT AVG(Salary) AS AverageSalary

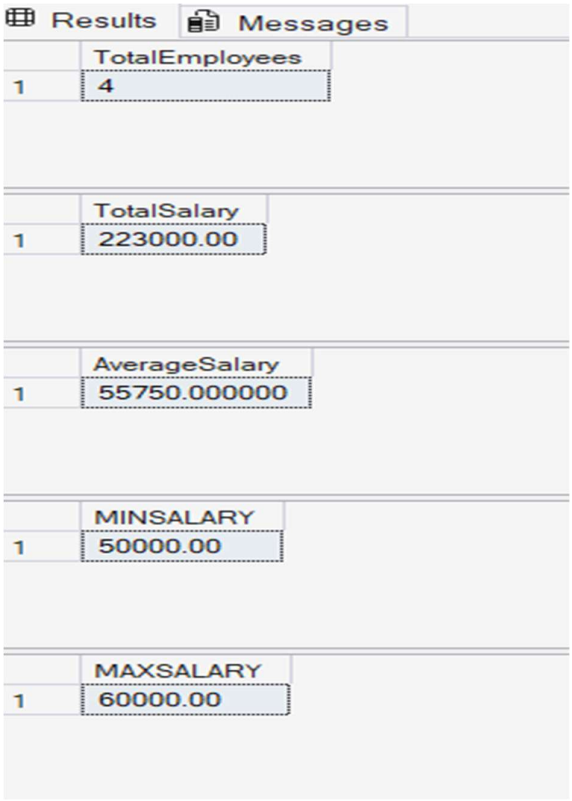
FROM Employees;

--min():to determine the minimum salary.

select min(salary) as MINSALARY from Employees

--MAX():to determine the maximum salary.

select MAX(salary) as MAXSALARY from Employees



--In Task 2, we explored how to refine and analyze data using SQL.

--The WHERE clause was used to filter records based on specific conditions, such as selecting employees from a particular department or those earning above a certain salary.

--Comparison operators (>, <, =, !=) allowed precise filtering.

--The ORDER BY clause helped sort records in ascending or descending order, making the data easier to interprete.

----------TASK 3----------

--1 group the records

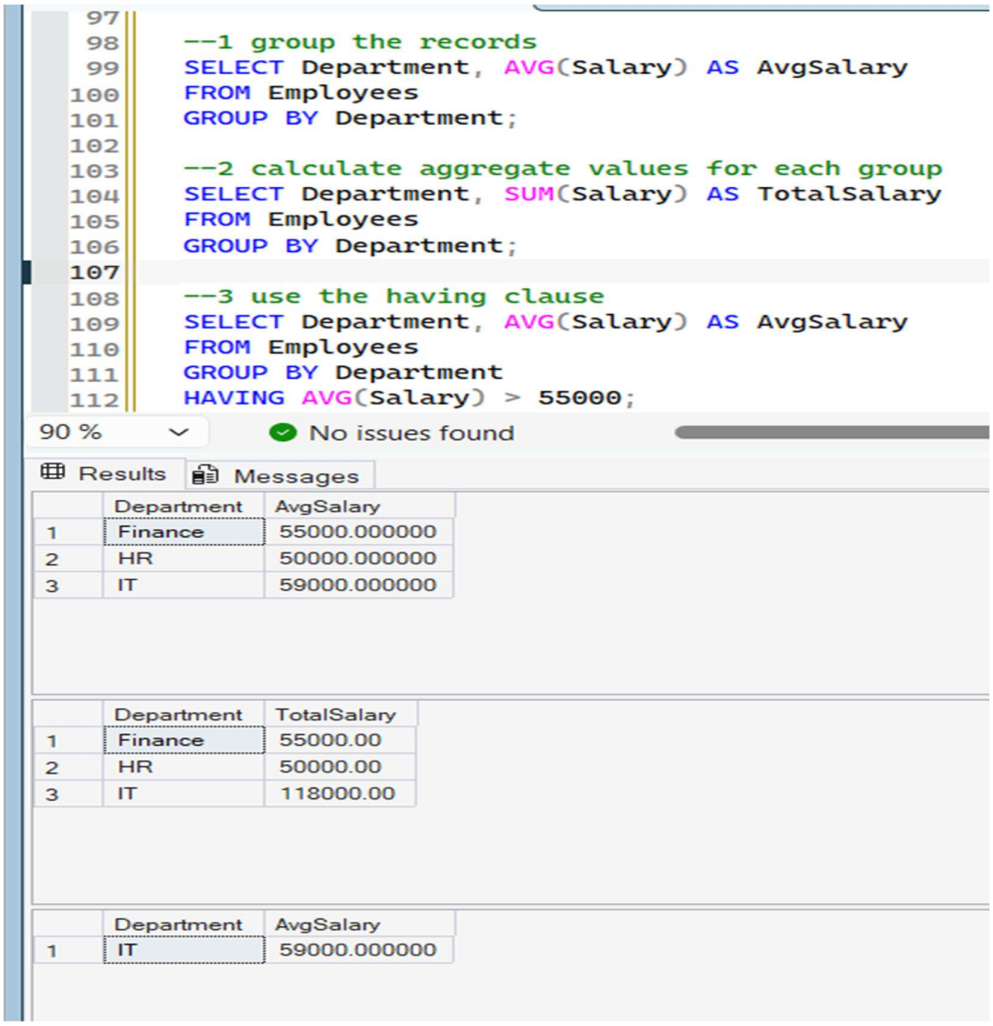
```
SELECT Department, AVG(Salary) AS AvgSalary
FROM Employees
GROUP BY Department;
```

--2 calculate aggregate values for each group

```
SELECT Department, SUM(Salary) AS TotalSalary
FROM Employees
GROUP BY Department;
```

--3 use the having clause

```
SELECT Department, AVG(Salary) AS AvgSalary
FROM Employees
GROUP BY Department
HAVING AVG(Salary) > 55000;
```



```sql
--1 group the records
SELECT Department, AVG(Salary) AS AvgSalary
FROM Employees
GROUP BY Department;

--2 calculate aggregate values for each group
SELECT Department, SUM(Salary) AS TotalSalary
FROM Employees
GROUP BY Department;

--3 use the having clause
SELECT Department, AVG(Salary) AS AvgSalary
FROM Employees
GROUP BY Department
HAVING AVG(Salary) > 55000;
```

90 % No issues found

Results Messages

	Department	AvgSalary
1	Finance	55000.000000
2	HR	50000.000000
3	IT	59000.000000

	Department	TotalSalary
1	Finance	55000.00
2	HR	50000.00
3	IT	118000.00

	Department	AvgSalary
1	IT	59000.000000

--task 3 shows how grouping and aggregation provide summarized insights,

--while the HAVING clause allows filtering of grouped results based on aggregate conditions.

----------**TASK 4**----------

--SQL JOINs are used to combine rows from two or more tables based on a related column.

--They allow us to query across multiple tables and see connected information.

--**CREATING THE TWO TABLES FOR JOINING**

```
CREATE TABLE Table1 (
   EmpID INT PRIMARY KEY,
   Name VARCHAR(50),
   Department VARCHAR(30)
);

INSERT INTO Table1 (EmpID, Name, Department)
VALUES
(101, 'Niklaus', 'HR'),
(102, 'Elijah', 'IT'),
(103, 'Rebekah', 'Finance'),
(104, 'Hayley', 'IT');


CREATE TABLE Table2 (
   EmpID INT PRIMARY KEY,
   Salary DECIMAL(10,2),
   Bonus DECIMAL(10,2)
);

INSERT INTO Table2 (EmpID, Salary, Bonus)
VALUES
(102, 60000, 7000),
```

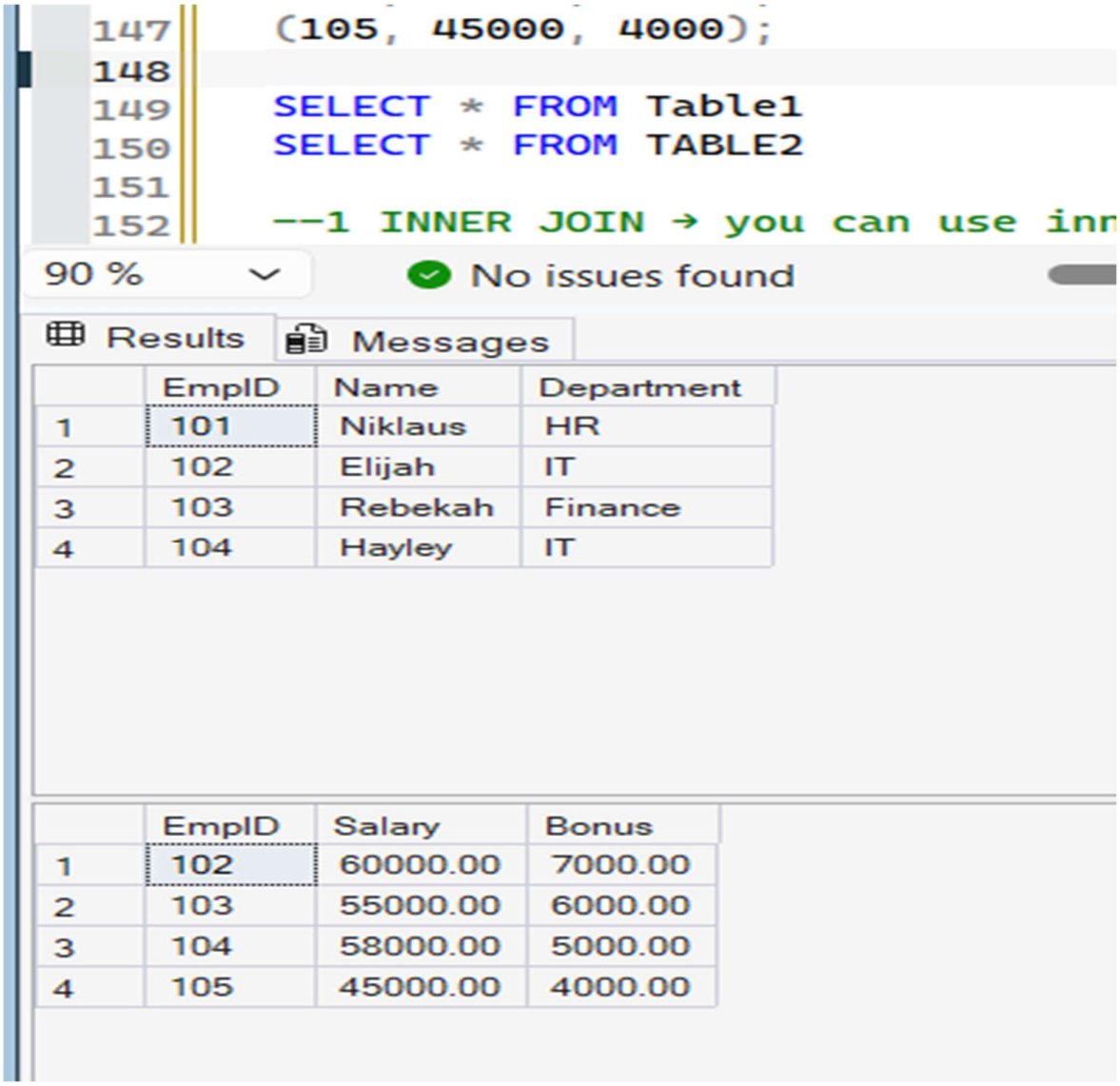
(103, 55000, 6000),

(104, 58000, 5000),

(105, 45000, 4000);

SELECT * FROM Table1

SELECT * FROM TABLE2



--**1 INNER JOIN** → you can use inner join if you want to combine rows from two or more tables based on related columns between them

SELECT T1.Empid,T1.name,T1.department,T2.salary,T2.bonus from TABLE1 T1 INNER JOIN TABLE2 T2 on T1.EmpID=T2.EmpID

```
152    --1 INNER JOIN → you can use inner join if you want to com
153    SELECT T1.Empid,T1.name,T1.department,T2.salary,T2.bonus
154    from
155    TABLE1 T1 INNER JOIN TABLE2 T2 on T1.EmpID=T2.EmpID
156
```

90 % No issues found

Results Messages

	Empid	name	department	salary	bonus
1	102	Elijah	IT	60000.00	7000.00
2	103	Rebekah	Finance	55000.00	6000.00
3	104	Hayley	IT	58000.00	5000.00

--**2 LEFT JOIN** → Returns all rows from the left table (Table1), and the matched rows from the right table (Table2). If no match exists, NULL values are shown for right table columns.

SELECT T1.Empid,T1.name,T1.department,T2.salary,T2.bonus from TABLE1 T1 LEFT JOIN TABLE2 T2 on T1.EmpID=T2.EmpID

```
156
157    --2 LEFT JOIN → Returns all rows from the left table (Table1),
158    SELECT T1.Empid,T1.name,T1.department,T2.salary,T2.bonus
159    from  TABLE1 T1 LEFT JOIN TABLE2 T2 on T1.EmpID=T2.EmpID
```

) % No issues found

Results Messages

Empid	name	department	salary	bonus
101	Niklaus	HR	NULL	NULL
102	Elijah	IT	60000.00	7000.00
103	Rebekah	Finance	55000.00	6000.00
104	Hayley	IT	58000.00	5000.00

--**3 RIGHT JOIN** → Returns all rows from the right table (Table2), and the matched rows from the left table (Table1). If no match exists, NULL values are shown for left table columns.

SELECT T1.Empid,T1.name,T1.department,T2.salary,T2.bonus from TABLE1 T1 RIGHT JOIN TABLE2 T2 on T1.EmpID=T2.EmpID

```
160
161     --3 RIGHT JOIN → Returns all rows from the right table (Table2), ar
162     SELECT T1.Empid,T1.name,T1.department,T2.salary,T2.bonus
163     from  TABLE1 T1 RIGHT JOIN TABLE2 T2 on T1.EmpID=T2.EmpID
164
165
```

90 % No issues found

Results Messages

	Empid	name	department	salary	bonus
1	102	Elijah	IT	60000.00	7000.00
2	103	Rebekah	Finance	55000.00	6000.00
3	104	Hayley	IT	58000.00	5000.00
4	NULL	NULL	NULL	45000.00	4000.00

ALL OUTPUT :

Results Messages

	Empid	name	department	salary	bonus
1	102	Elijah	IT	60000.00	7000.00
2	103	Rebekah	Finance	55000.00	6000.00
3	104	Hayley	IT	58000.00	5000.00

	Empid	name	department	salary	bonus
1	101	Niklaus	HR	NULL	NULL
2	102	Elijah	IT	60000.00	7000.00
3	103	Rebekah	Finance	55000.00	6000.00
4	104	Hayley	IT	58000.00	5000.00

	Empid	name	department	salary	bonus
1	102	Elijah	IT	60000.00	7000.00
2	103	Rebekah	Finance	55000.00	6000.00
3	104	Hayley	IT	58000.00	5000.00
4	NULL	NULL	NULL	45000.00	4000.00

----------TASK 5----------

--A subquery is a query nested inside another SQL query. It is often used in the WHERE, FROM, or SELECT clauses to perform intermediate calculations or filtering.

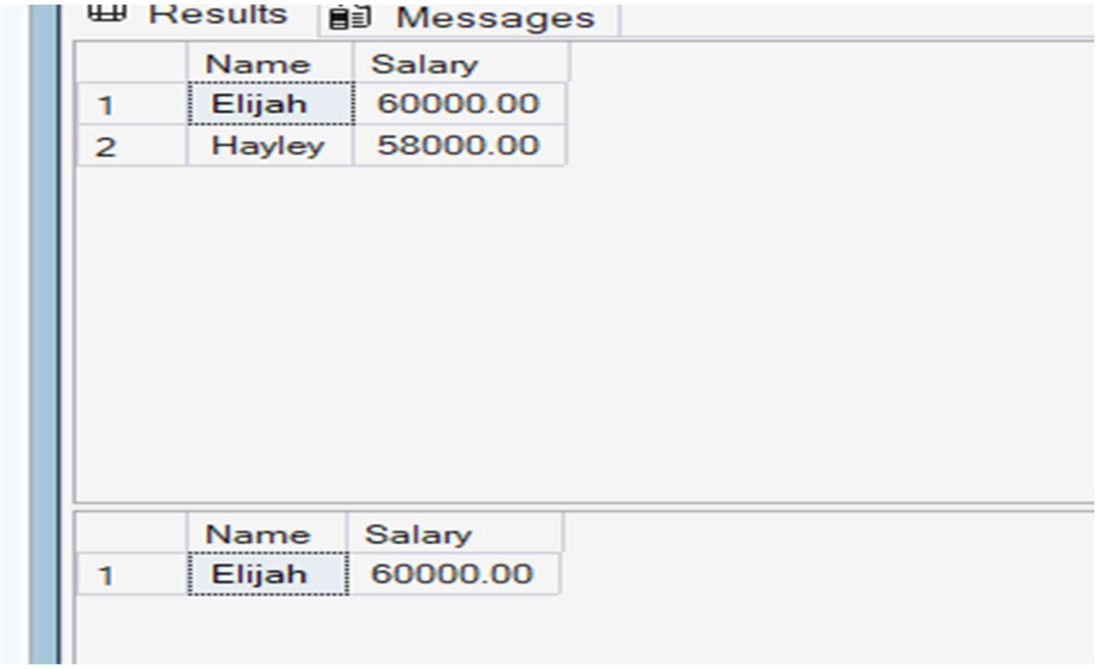
--Subqueries allow us to compare values against aggregated results or dynamically generated sets.

--1 Find employees earning more than the average salary.

```
SELECT Name, Salary
FROM Employees
WHERE Salary > (SELECT AVG(Salary) FROM Employees);
```

--**2 Find employees whose salary equals the maximum salary**

```
SELECT Name, Salary
FROM Employees
WHERE Salary = (SELECT MAX(Salary) FROM Employees);
```



EXCEL

----------TASK 6----------

1. Formatting

- **Bold and centre the header row.**
- **Applied a background color to headers (blue).**
- **Formatted Salary as currency.**
- **Formatted JoiningDate as short date.**

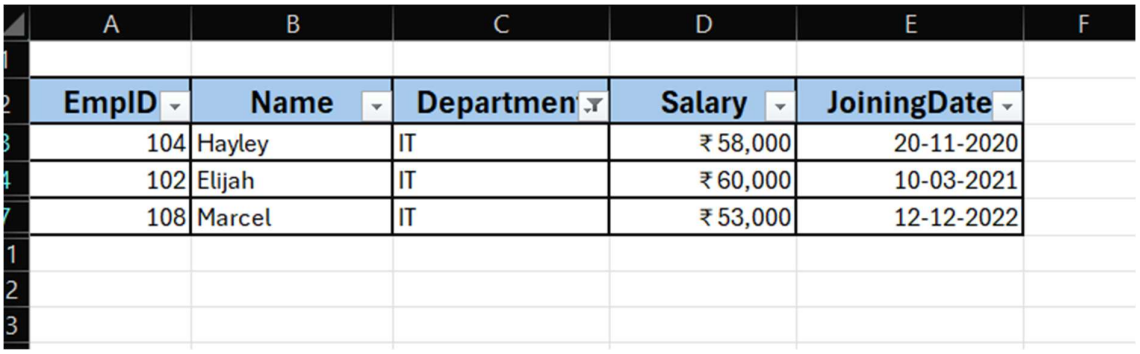
EmpID	Name	Department	Salary	JoiningDate
101	Niklaus	HR	₹ 50,000	15-01-2022
102	Elijah	IT	₹ 60,000	10-03-2021
103	Rebekah	Finance	₹ 55,000	01-07-2023
104	Hayley	IT	₹ 58,000	20-11-2020
105	Kol	HR	₹ 52,000	05-05-2024
106	Freya	Marketing	₹ 47,000	14-02-2023
107	Davina	Finance	₹ 62,000	30-09-2021
108	Marcel	IT	₹ 53,000	12-12-2022

2. Sortted by JoiningDate (Ascending) → earliest hires first.

EmpID	Name	Department	Salary	JoiningDate
104	Hayley	IT	₹ 58,000	20-11-2020
102	Elijah	IT	₹ 60,000	10-03-2021
107	Davina	Finance	₹ 62,000	30-09-2021
101	Niklaus	HR	₹ 50,000	15-01-2022
108	Marcel	IT	₹ 53,000	12-12-2022
106	Freya	Marketing	₹ 47,000	14-02-2023
103	Rebekah	Finance	₹ 55,000	01-07-2023
105	Kol	HR	₹ 52,000	05-05-2024

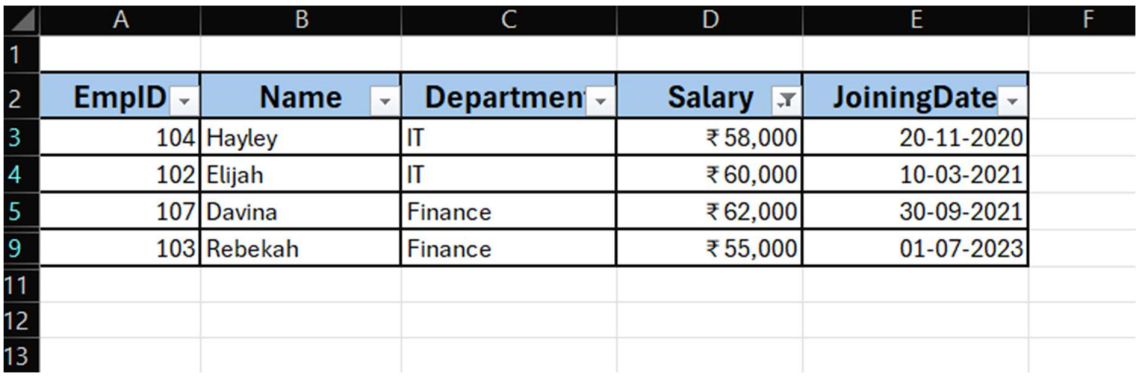
3. Filtering

- **Show only IT Department employees.**



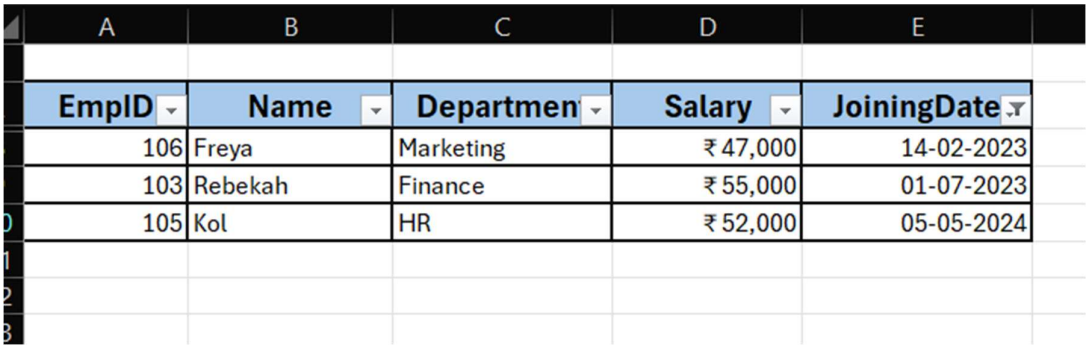
EmpID	Name	Departmen	Salary	JoiningDate
104	Hayley	IT	₹ 58,000	20-11-2020
102	Elijah	IT	₹ 60,000	10-03-2021
108	Marcel	IT	₹ 53,000	12-12-2022

- **Show employees with Salary > 55,000.**



EmpID	Name	Departmen	Salary	JoiningDate
104	Hayley	IT	₹ 58,000	20-11-2020
102	Elijah	IT	₹ 60,000	10-03-2021
107	Davina	Finance	₹ 62,000	30-09-2021
103	Rebekah	Finance	₹ 55,000	01-07-2023

- **Show employees who joined after 01-Jan-2023.**

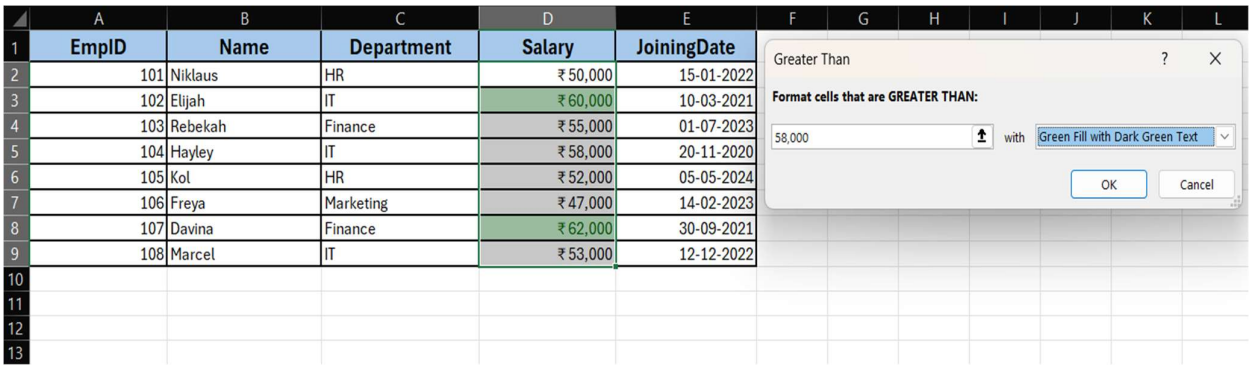


EmpID	Name	Departmen	Salary	JoiningDate
106	Freya	Marketing	₹ 47,000	14-02-2023
103	Rebekah	Finance	₹ 55,000	01-07-2023
105	Kol	HR	₹ 52,000	05-05-2024

----------TASK7----------

1.Conditional formatting

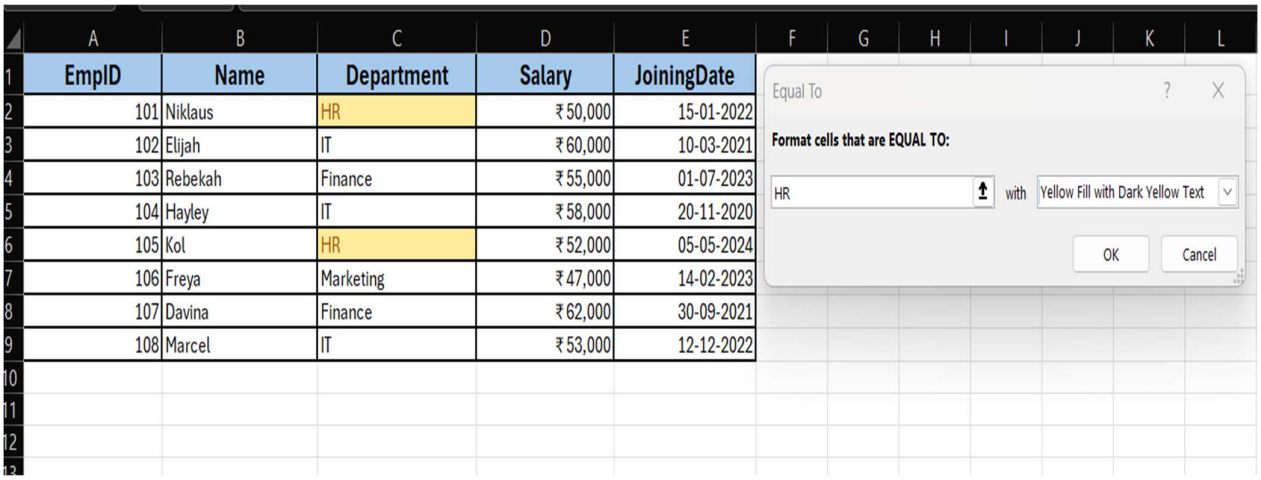
- **Highlight the cell with salary greater than 58,000.**



Output:

EmpID	Name	Department	Salary	JoiningDate
101	Niklaus	HR	₹50,000	15-01-2022
102	Elijah	IT	₹60,000	10-03-2021
103	Rebekah	Finance	₹55,000	01-07-2023
104	Hayley	IT	₹58,000	20-11-2020
105	Kol	HR	₹52,000	05-05-2024
106	Freya	Marketing	₹47,000	14-02-2023
107	Davina	Finance	₹62,000	30-09-2021
108	Marcel	IT	₹53,000	12-12-2022

- **Highlight the cell if the department is HR**



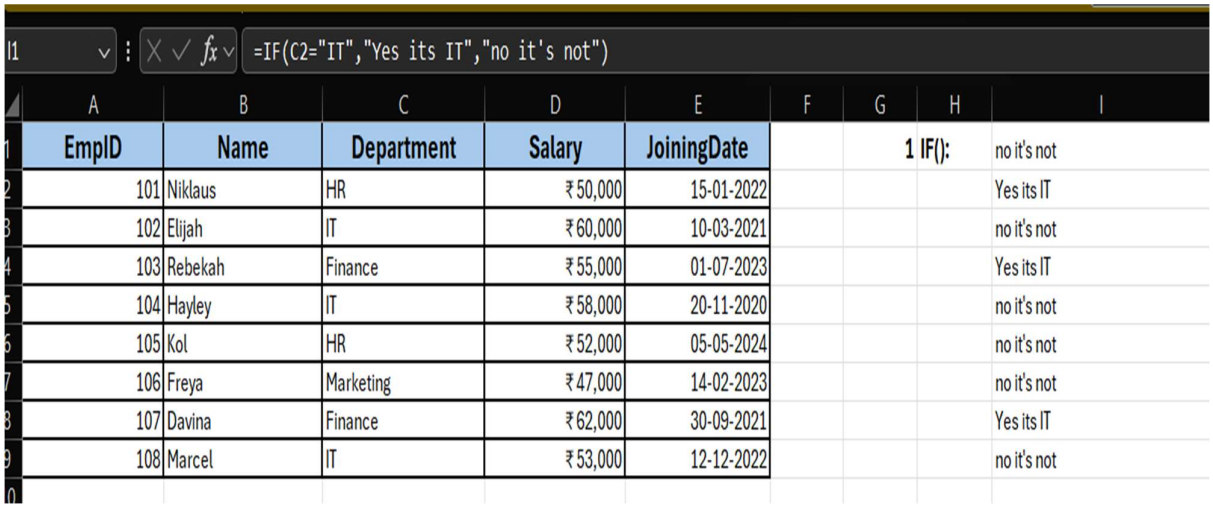
Output:

EmpID	Name	Department	Salary	JoiningDate
101	Niklaus	HR	₹50,000	15-01-2022
102	Elijah	IT	₹60,000	10-03-2021
103	Rebekah	Finance	₹55,000	01-07-2023
104	Hayley	IT	₹58,000	20-11-2020
105	Kol	HR	₹52,000	05-05-2024
106	Freya	Marketing	₹47,000	14-02-2023
107	Davina	Finance	₹62,000	30-09-2021
108	Marcel	IT	₹53,000	12-12-2022

- **Conditional Formatting in Excel is a feature that automatically changes the appearance of cells based on rules you define. It's like telling Excel: *"If this condition is true, highlight the cell."***

2.Excel Functions

- **If():The IF function in Excel is used to test a condition and return one value if the condition is TRUE, and another if it is FALSE.**

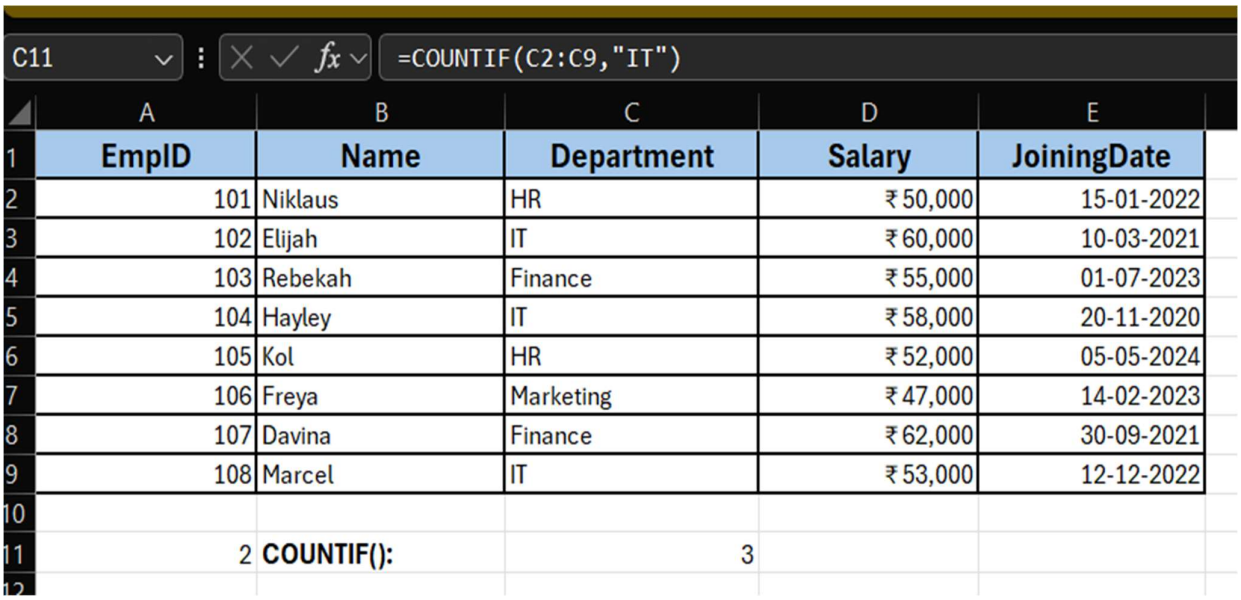


Explanation:

- **C2="IT" → checks if the value in cell C2 is exactly "IT".**
- **If TRUE → Excel shows Yes it's IT.**
- **If FALSE → Excel shows No it's not.**

- **COUNTIF():The COUNTIF function counts the number of cells in a range that meet a specific condition.**

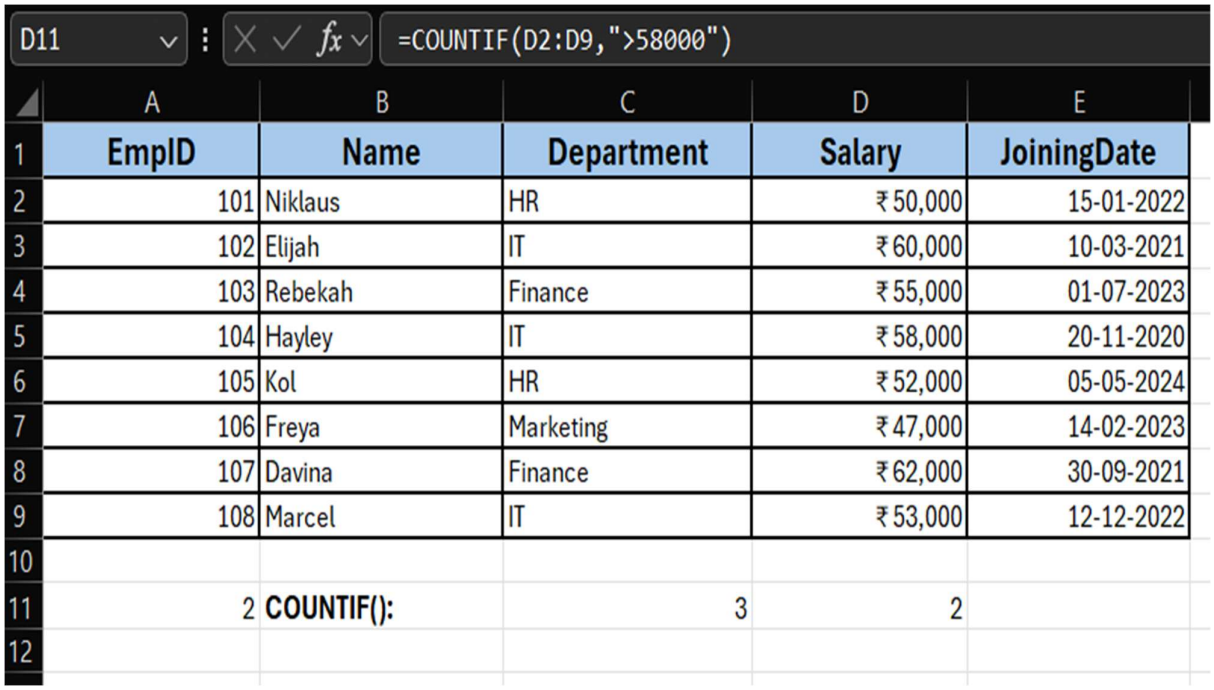
- **EG 1: count if department is IT**



C11 =COUNTIF(C2:C9,"IT")

	A	B	C	D	E
1	EmpID	Name	Department	Salary	JoiningDate
2	101	Niklaus	HR	₹50,000	15-01-2022
3	102	Elijah	IT	₹60,000	10-03-2021
4	103	Rebekah	Finance	₹55,000	01-07-2023
5	104	Hayley	IT	₹58,000	20-11-2020
6	105	Kol	HR	₹52,000	05-05-2024
7	106	Freya	Marketing	₹47,000	14-02-2023
8	107	Davina	Finance	₹62,000	30-09-2021
9	108	Marcel	IT	₹53,000	12-12-2022
10					
11	2	**COUNTIF():**	3		

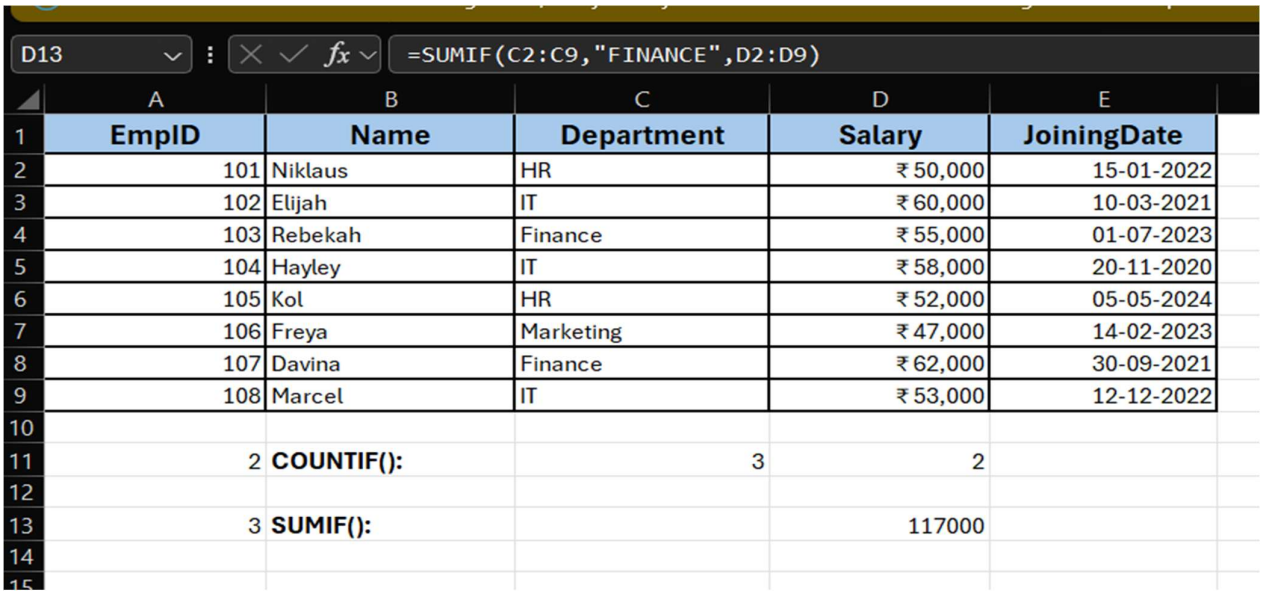
- **EG 2: count if salary is greater than 58,000**



D11 =COUNTIF(D2:D9,">58000")

	A	B	C	D	E
1	EmpID	Name	Department	Salary	JoiningDate
2	101	Niklaus	HR	₹50,000	15-01-2022
3	102	Elijah	IT	₹60,000	10-03-2021
4	103	Rebekah	Finance	₹55,000	01-07-2023
5	104	Hayley	IT	₹58,000	20-11-2020
6	105	Kol	HR	₹52,000	05-05-2024
7	106	Freya	Marketing	₹47,000	14-02-2023
8	107	Davina	Finance	₹62,000	30-09-2021
9	108	Marcel	IT	₹53,000	12-12-2022
10					
11	2	**COUNTIF():**	3	2	
12					

- **Sumif(): The SUMIF function adds up values in a range that meet a specific condition. It's like saying: *"Sum only those numbers that match my rule."***
- **Eg: find the total salary of finance department**



D13 | =SUMIF(C2:C9,"FINANCE",D2:D9)

	A	B	C	D	E
1	EmpID	Name	Department	Salary	JoiningDate
2	101	Niklaus	HR	₹ 50,000	15-01-2022
3	102	Elijah	IT	₹ 60,000	10-03-2021
4	103	Rebekah	Finance	₹ 55,000	01-07-2023
5	104	Hayley	IT	₹ 58,000	20-11-2020
6	105	Kol	HR	₹ 52,000	05-05-2024
7	106	Freya	Marketing	₹ 47,000	14-02-2023
8	107	Davina	Finance	₹ 62,000	30-09-2021
9	108	Marcel	IT	₹ 53,000	12-12-2022
10					
11	2	**COUNTIF():**	3	2	
12					
13	3	**SUMIF():**		117000	
14					

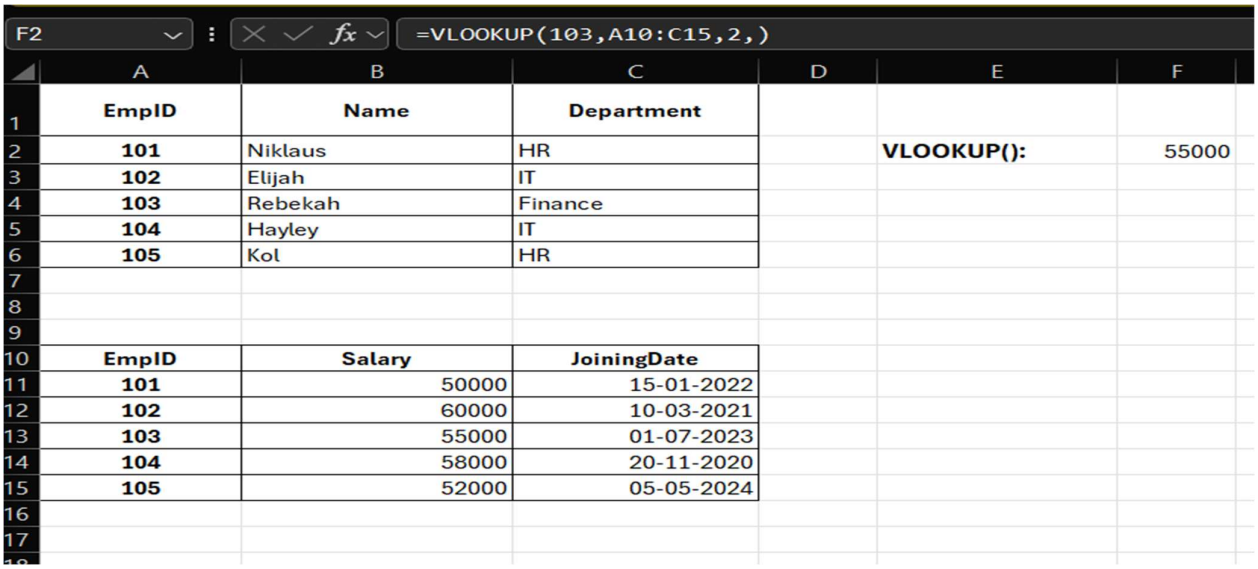
----------TASK 8----------

TABLES USED FOR LOOKUP FUNCTION TASK

	A	B	C	D
1	**EmpID**	**Name**	**Department**	
2	**101**	Niklaus	HR	
3	**102**	Elijah	IT	
4	**103**	Rebekah	Finance	
5	**104**	Hayley	IT	
6	**105**	Kol	HR	
7				
8				
9				
10	**EmpID**	**Salary**	**JoiningDate**	
11	**101**	50000	15-01-2022	
12	**102**	60000	10-03-2021	
13	**103**	55000	01-07-2023	
14	**104**	58000	20-11-2020	
15	**105**	52000	05-05-2024	
16				
17				

VLOOKUP(): The VLOOKUP() function in Excel is used to search for a value in the first column of a table and return a value from another column in the same row. It stands for Vertical Lookup because it looks down a column to find the match.

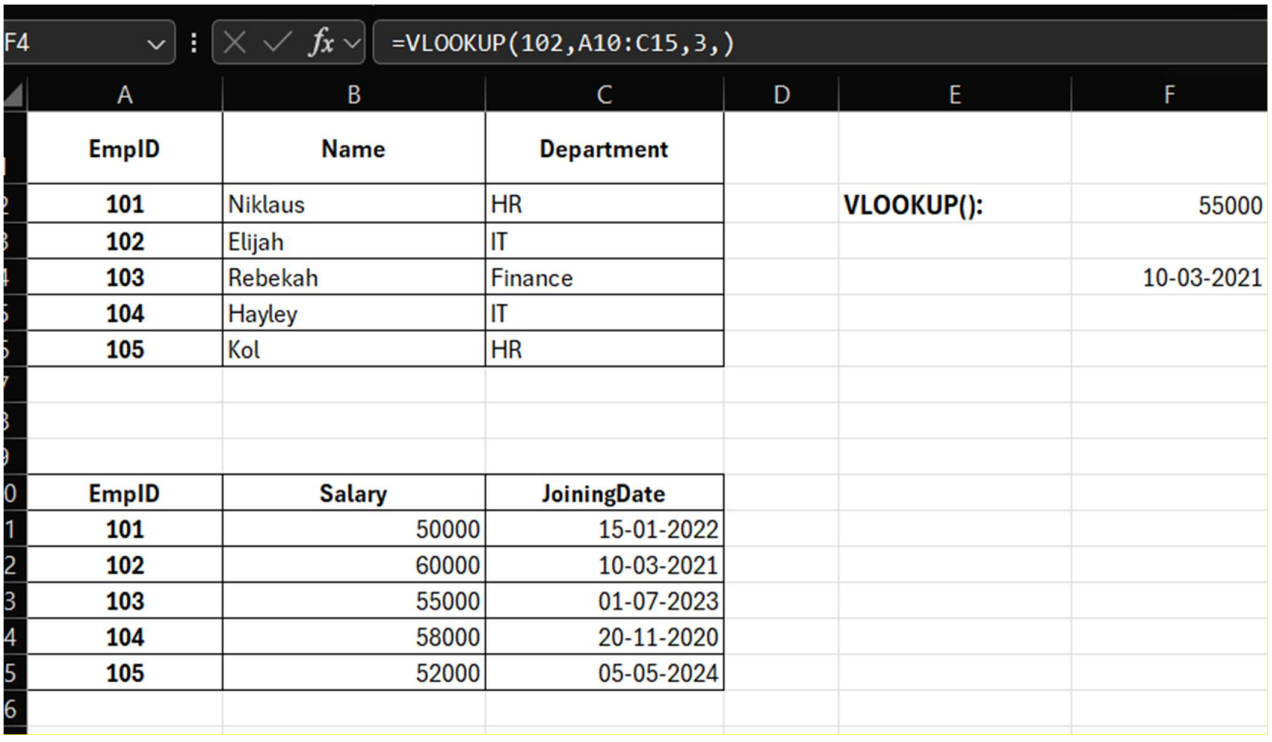
- **Find the Salary of EmpID = 103 (Rebekah)**



F2 =VLOOKUP(103,A10:C15,2,)

	A	B	C	D	E	F
1	**EmpID**	**Name**	**Department**			
2	**101**	Niklaus	HR		**VLOOKUP():**	55000
3	**102**	Elijah	IT			
4	**103**	Rebekah	Finance			
5	**104**	Hayley	IT			
6	**105**	Kol	HR			
7						
8						
9						
10	**EmpID**	**Salary**	**JoiningDate**			
11	**101**	50000	15-01-2022			
12	**102**	60000	10-03-2021			
13	**103**	55000	01-07-2023			
14	**104**	58000	20-11-2020			
15	**105**	52000	05-05-2024			
16						
17						

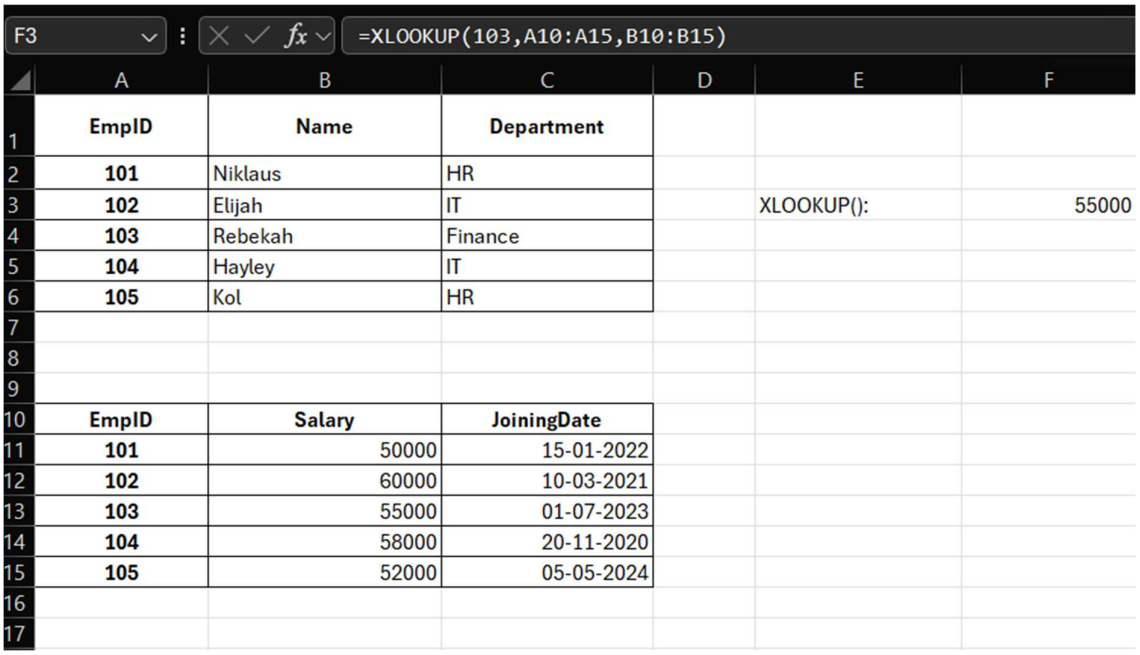
- **Find Elijah's joining date:**



F4 =VLOOKUP(102,A10:C15,3,)

	A	B	C	D	E	F
	EmpID	**Name**	**Department**			
2	**101**	Niklaus	HR		**VLOOKUP():**	55000
3	**102**	Elijah	IT			
4	**103**	Rebekah	Finance			10-03-2021
5	**104**	Hayley	IT			
6	**105**	Kol	HR			
7						
8						
9						
10	**EmpID**	**Salary**	**JoiningDate**			
11	**101**	50000	15-01-2022			
12	**102**	60000	10-03-2021			
13	**103**	55000	01-07-2023			
14	**104**	58000	20-11-2020			
15	**105**	52000	05-05-2024			
16						

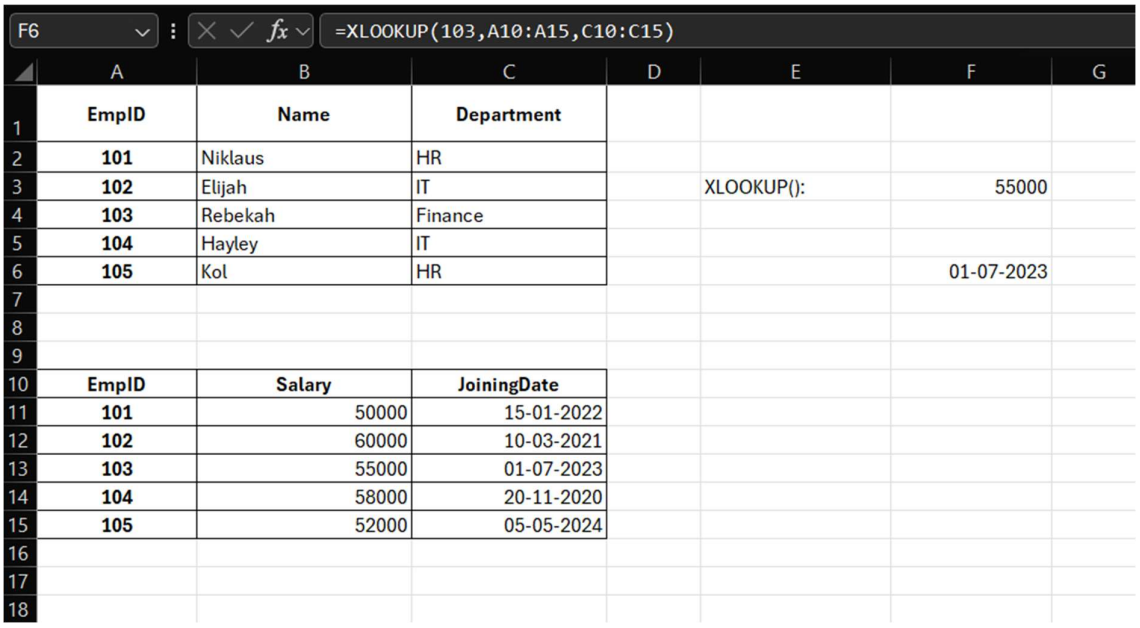
- **XLOOKUP():The XLOOKUP() function in Excel is used to search for a value in one column (lookup array) and return a corresponding value from another column (return array). It's more powerful and flexible than VLOOKUP because it doesn't depend on column positions.**
    - **Find the Salary of EmpID = 103**



F3 =XLOOKUP(103,A10:A15,B10:B15)

	A	B	C	D	E	F
1	**EmpID**	**Name**	**Department**			
2	**101**	Niklaus	HR			
3	**102**	Elijah	IT		XLOOKUP():	55000
4	**103**	Rebekah	Finance			
5	**104**	Hayley	IT			
6	**105**	Kol	HR			
7						
8						
9						
10	**EmpID**	**Salary**	**JoiningDate**			
11	**101**	50000	15-01-2022			
12	**102**	60000	10-03-2021			
13	**103**	55000	01-07-2023			
14	**104**	58000	20-11-2020			
15	**105**	52000	05-05-2024			
16						
17						

- **EMPLOYEE 103 JOINING DATE**



F6 =XLOOKUP(103,A10:A15,C10:C15)

	A	B	C	D	E	F	G
1	**EmpID**	**Name**	**Department**				
2	**101**	Niklaus	HR				
3	**102**	Elijah	IT		XLOOKUP():	55000	
4	**103**	Rebekah	Finance				
5	**104**	Hayley	IT				
6	**105**	Kol	HR			01-07-2023	
7							
8							
9							
10	**EmpID**	**Salary**	**JoiningDate**				
11	**101**	50000	15-01-2022				
12	**102**	60000	10-03-2021				
13	**103**	55000	01-07-2023				
14	**104**	58000	20-11-2020				
15	**105**	52000	05-05-2024				
16							
17							
18							

----------TASK 9----------

1.What is pivot table?

A Pivot Table is one of Excel's most powerful tools. It allows you to:

- **Summarize large datasets quickly.**
- **Rearrange (pivot) data to view it from different perspectives.**
- **Perform calculations like Sum, Count, Average without writing formulas.**
- **Provide a structured summary that can be linked directly to charts**

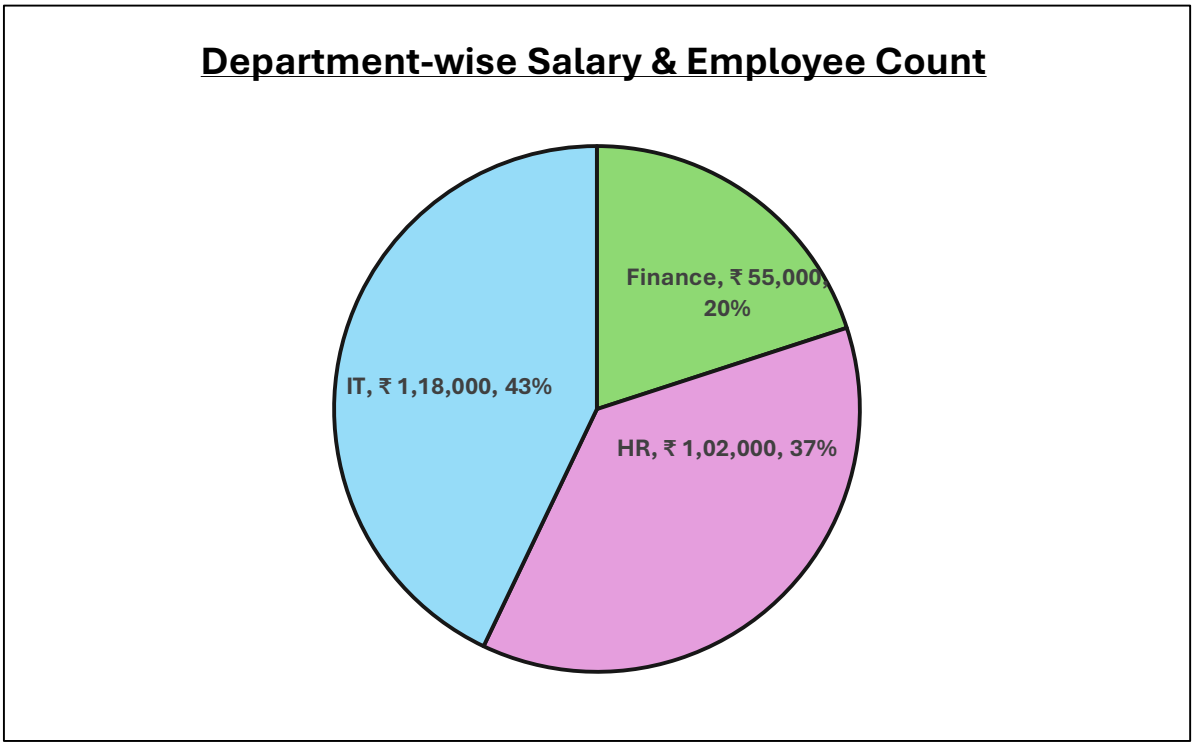
DATA USED:

EmpID	Name	Department	Salary
101	Niklaus	HR	50000
102	Elijah	IT	60000
103	Rebekah	Finance	55000
104	Hayley	IT	58000
105	Kol	HR	52000

2.Built the pivot table

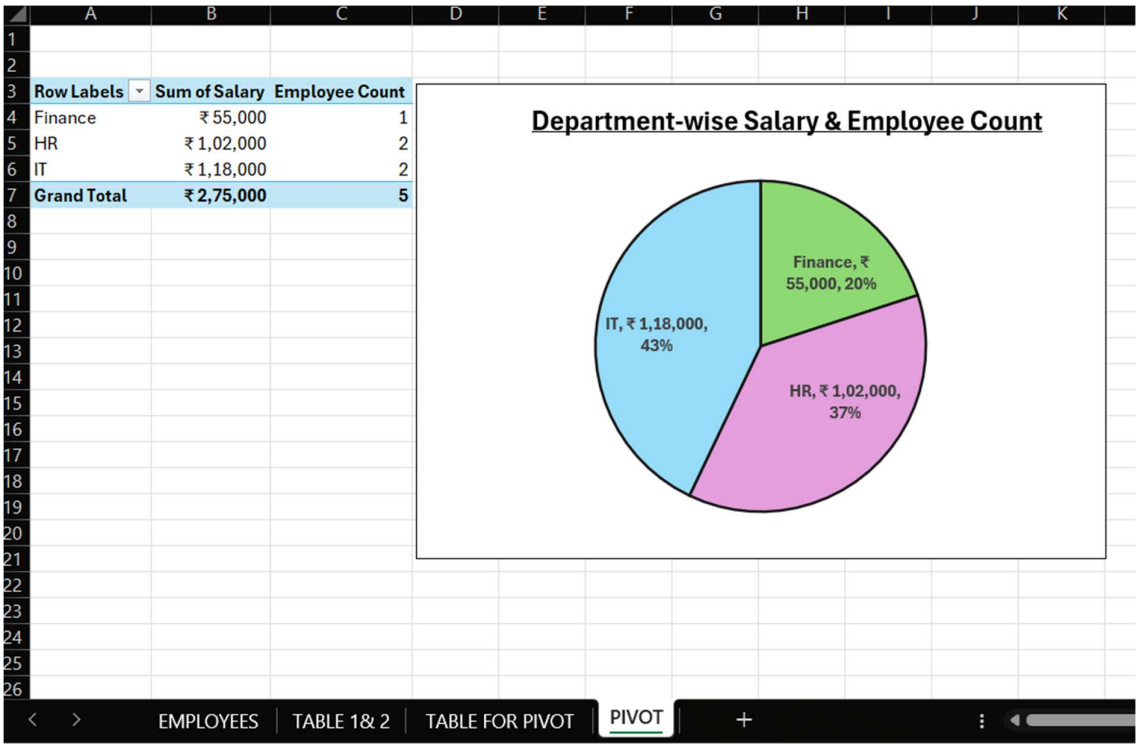
	A	B	C
1			
2			
3	**Row Labels**	**Sum of Salary**	**Employee Count**
4	Finance	₹ 55,000	1
5	HR	₹ 1,02,000	2
6	IT	₹ 1,18,000	2
7	**Grand Total**	**₹ 2,75,000**	**5**
8			
9			
10			

3.Create a chart and analyze the data



- **The chart visually represents how salaries are distributed across departments.**
- **IT department has the highest payroll cost (43%).**
- **HR department follows with 37%.**
- **Finance department has the smallest share (20%).**
- **This helps management quickly identify which department consumes the largest portion of salary expenses.**

4.Analyze the chart



Row Labels	Sum of Salary	Employee Count
Finance	₹ 55,000	1
HR	₹ 1,02,000	2
IT	₹ 1,18,000	2
Grand Total	**₹ 2,75,000**	**5**

NOTE:

1.OUTPUTS OF EVERY QUERY IS SCREENSHOTTED AND PASTED BELOW EVERY QUERY AND EXCEL OUTPUT IS PROVIDED WITH EXPLANATION AND SCREENSHOTS OF OUTPUT

2.ALL TABLES USED IN EXCEL WORKBOOK ARE ATTACHED IN THE DRIVE AND SQL QUERY IS ALSO ATTCHED IN TEXT FORMAT.